Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

TION TECHNOLOGY
nology

Methods of Research (DCIT 60)

Assignment 4: Formulating the Title and Research Proposal

ANSWER SHEET

Proponents: Group Leader: Casino, Lanz Michael Group Members: Casaverde, Mike Sampaga, Charls
Proposed Title (Write 3 possible titles for your proposal): -Development of a Web-Based Record Management System for Cavite State University's Extension Office -Implementation of Web-Based Record Management System for Extension Office in Cavite State University -Development of an Integrated Web-Based Record Management System for Cavite State University's Extension Office Operations
Sustainable Development Goals: SDGS 17: Partnerships for the Goals

CvSU Research Thematic Area

Smart Engineering, ICT, and Industrial Competitiveness

Rationale

Cavite State University (CvSU) stands as a premier institution committed to academic
ension Office plays a vital role in
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of the activities. Currently, the
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database. However, as the scope and scale of extension activities have expanded over the years, the limitations of this conventional system have become increasingly apparent.

The growing problem of extension offices, coupled with the increasing demands for transparency, accountability, and quick access to information, necessitates a more robust and efficient approach to managing the office's documentary requirements. The volume of records continues to increase annually as the university expands its community outreach. The current record management system in the CvSU Extension Office faces several critical challenges that hinder operational efficiency. Document retrieval is time-consuming, often requiring staff members to manually search Google Drive, which can take considerable time, especially when urgent information is needed. The risk of document loss or misplacement is heightened due to the physical nature of storage and lack of centralized digital backup systems. While generic document management systems and cloud storage solutions exist, they lack the specialized features required for extension office operations, such as program tracking, monitoring, and evaluation reports. Some commercial systems often come with high licensing costs and may not be customizable to fit the unique needs and processes of the CvSU Extension Office. Furthermore, the existing system makes collaboration and information sharing among extension staff difficult, particularly when multiple personnel need to access the same documents simultaneously or when working remotely. These inefficiencies translate to productivity losses, increased administrative burden on staff, potential compliance issues with university documentation requirements, and delayed response time to inquiries.

The development of a web-based record management system addresses an urgent and pressing need for the CvSU Extension Office. With the increasing emphasis on digital transformation across educational institutions, particularly accelerated by recent global shifts toward online operations, modernizing record management practices is no longer optional but essential for institutional competitiveness and sustainability. A web-based system offers distinct advantages over traditional methods, such as centralized storage with secure access controls. real-time accessibility from any location with internet connectivity, automated backup and disaster recovery capabilities, streamlined workflows for document approval and processing, and robust search and retrieval functions. These features directly address the identified problems while positioning the Extension Office to better serve the community.

Significance of the Study

This study is significant as it addresses the critical need for efficient and systematic record management in university extension services. With extension offices managing numerous community programs, partnerships, and activities, a web-based system will streamline operations and enhance service delivery. The Cavite State University's Extension Office will modernize information handling and enhance the overall effectiveness of extension operations.

CvSU Office of the Vice President for Research, Innovation, and Extension. The OVPRIE has many tasks daily, the web-based system will streamline their daily administrative tasks by providing centralized access to extension records to reduce time spent on document retrieval. Also, it will eliminate manual filing processes of the staff so that they can focus more on core extension activities and community engagement rather than administrative burdens.

Extension Office Administrators. The system will provide real-time access to comprehensive reports and analytics. It can also support the administration in decision-making for program planning and resource allocation. It will also facilitate easier monitoring of extension projects, activities timelines, and performance came from different coordinators across different colleges. The administrators can easily manage the user roles by updating the accounts and information about it.

College and Department Coordinators. The system allows them to input the details of the extension activities. In this manner, it will improve record management to support institutional compliance requirements, accreditation process, and transparent reporting of extension activities. This system is significant for them since it eliminates creating reports manually and it will give them the opportunity to send the report on time.

Future Researchers and Developers. This study will serve as a reference for developing similar record management systems in other university offices or higher education institutions.

Objectives of the Study

This study aims to allow the Cavite State University's Extension Office to organize their documentation using a Web-Based Information Management System. In addition, the system helps to have a secure and proper document storage or database.

- To design a web application using the requirements gathered based on the inputs from the clients gathered using survey and interview.
- To allow the coordinators to create, edit, delete and view reports of extension office activities and training.
- Support real-time data access for the coordinators and admins.
- Generate an accurate summary or report analysis using graphs.

Literature Review

Record management is an essential tool for organized records, data, and files that an organization has. Records are essential to academic institutions, documenting actions and transactions while providing reliable evidence of institutional decisions (Sebucao-Orias, 2023). This topic deals with record management of extension offices developing a Web-Based Management Information System for enhancing the management of extension services in Cavite State University, specifically the Office of the Vice President for Research, Innovation, and Extension. Currently, the identified problems are the absence of a centralized database for storing and organizing all documents, difficulties in fetching and submitting required documents due to manual processes, and the lack of a dedicated system specifically designed for extension service operations. This topic aims to resolve the handling of documents and files of Cavite State University's Extension Services in terms of having a centralized database. This system is designed to replace traditional, paper-based processes with an automated digital workflow, a core shift discussed in literature concerning digital libraries and record management (Urmeneta, 2025).

The system's function is revolved in the CRUD functionality: creating digitized forms or direct input, editing the documents, deleting, secure storage, and efficient retrieval. Automating these processes is a critical function, leveraging technologies to streamline operations. This includes automated logging-in and user authentication, digital workflow automation for document approval and tracking, and record-keeping to maintain an accurate history of extension activities, a feature common to effective tertiary institution record systems (Uka, 2019). By automating these processes, the web-based record management system enhances organizational accountability, reduces human error, and ensures the immediate generation of institutional reports, reflecting the broader goal of automating data management in academic settings (Sebucao-Orias, 2023 & Nurhidayah et. al, 2023).

In the study of Ismael and Okumus (2017), the developers split their system into three parts: document management, document storage, and document retrieval and sharing. They worked with both external paper documents that needed scanning and internal documents created in the system, storing everything as PDF files on separate servers. This method of management of recorded documents significantly shows effectiveness and sufficiency in terms of handing documents whether it is online or paper. While on the study of Rosario et al. (2016), it focuses on handling thesis-related files like proposals, papers, and teaser videos, using tags to make searching easier but it is still a record management system related because it involves storing and retrieval of documents. The developers in Rosario et al. (2016) made document management just one part of their larger thesis portal, focusing on three things: storing, indexing, and search and retrieval. Their system handled thesis-related files like proposals, papers, and teaser videos, using tags to make searching easier.

Technologically wise, this system will employ technical tools which are accessible and sufficient web technologies for this proposed record management system such as React.js, PHP for backend, Node.js for the backend, and mongoDB and Firebase for database management. In addition, it will also use programming languages such as HTML, CSS, and JavaScript for front end. These technologies will be utilized in this system in order to develop the functional record management system and fulfill the goals and purpose of it with main features including document upload and retrieval, user access controls, automated reporting, and cloud storage integration are achievable using standard CRUD operations and established libraries, representing low to medium complexity levels. The system will be feasible in many months of developing, prototyping, testing, and piloting. After that, the system will have a tutorial session for users and implement the system in the University afterwards. In the study of Urmeneta et al. (2025), it shows that they employed XAMPP as their local web server environment, incorporating Apache, MySQL, PHP, and Perl, alongside Bootstrap and Visual Studio Code (VSCode) as the code editor to guarantee system reliability and security. ApexCharts.js was implemented for data visualization to improve the overall presentation of data. The developers utilized sets of programs in order to develop a

Web-Based Extension Services Record Management System for the Universities and Colleges that the developers have chosen.

Methodology

The development approach for this system will employ the Agile methodology because it encourages rapid adaptability to changes in requirements and takes clients' feedback into consideration throughout the development process; this technique works well for this system. Additionally, the developers chose this approach because it allows the team to work in short cycles called sprints, making it easier to adapt to feedback and changes throughout the project.

This system will have six development phases which are planning, analysis, design, testing, implementation, and maintenance and support phase. The development operation will begin the planning phase, where the developers will seek requirements through interviews, surveys, and consultations with the OVPRIE staff, administrators, and college coordinators.The particular requirements, difficulties, and expectations of the end users in relation to the record management system will be determined at this phase. After gathering the requirements, the developers will proceed to the analysis stage, where they will review the data and decide on the features and structure of the system.

Subsequently, designing the system will be the next phase. During this phase, the developers will create the system architecture, database structure, and user interface designs based on the analyzed requirements. In addition, it involves selecting the right frameworks, technologies, and tools for the development of the system. Before beginning real development, clients will be shown the design prototypes in order to get their input and make sure the suggested design satisfies their needs.

Regular meetings and demonstrations with stakeholders will be held throughout the development process to update them on developments, get their feedback, and make any required adjustments. The final system will be in line with the real requirements of the extension office and its users according to this productive approach. In order to maintain the system's reliability and quality, testing will be done regularly during each sprint to find and fix problems early. In order to keep the system current and functional, the Agile methodology also allows the addition of new features or changes depending on input from clients in the proposed project. Before being deployed and implemented at Cavite State University's Extension Office, the system will go through final testing and assessment after all sprints are finished.

Expected Output

In this study, the developers are expecting to develop a functional web-based record management system specifically designed for Cavite State University's Extension Office. The system will use a centralized database that stores and organizes all extension-related documents and records, particularly in program documentation, activity reports, partnership agreements, and program effectiveness from various college coordinators. In addition, the system will feature user-friendly interfaces solely for administrators, staff members, and college coordinators, handling all the documents regarding extension programs with each role having appropriate access levels and functionalities.

Additionally, the system will provide a comprehensive user-interface for inputting documents and needed information as the administrators requirements. This includes the transformation of the current way of submitting reports through Excel spreadsheets and using other different formats into an integrated and standardized method of submitting records directly within the system that has a specified user interface with it. Additionally, it will be a useful resource for scholars and organizations in the future that want to use similar digital solutions to administer extension services, which will further enhance information management practices in higher education.

When it comes to integrated web-based systems, the developers expect to have a functional web-based record management system that administrators, staff, and college coordinators of Cavite State University can employ over the web and that eases the handling and participation in any extension services program. The system intends to serve all stakeholders in all of their daily functions and to equip them with the systems and functions that allow them to manage the extension records and all of their derivatives efficiently and effectively from the creation of the records to their archiving. The system will enable administrators to generate all-inclusive reports, gain insight into all extension activities, and make valuable real-time analyses and decisions. College coordinators will gain portal access to submit reports of their activities, monitor their reports, and request and retrieve documents and templates corresponding to the administrative requirements. Since the system will allow all users to perform their specific functions and responsibilities, the system will promote uniformity of functions in the extension office and integrated systems of record management across the university.

Multiple Constraints

The development of the Web-Based Record Management System for Extension Office in CvSU Main must follow various constraints that will influence its design and implementation. These constraints include technical, functional, user, and budget and resource constraints that will guide the development of the system across the main campus and 12 satellite campuses.

Technical Constraints

The system must be web-based only, not a mobile app, and it must work smoothly on browsers like Chrome, Firefox, Brave, etc. Internet connectivity issues may interfere with system consistency and performance since the system requires stable internet connection for its integrated web-based design across all university campuses and colleges. The system must also handle simultaneous users since it will be used by the main campus and 12 satellite campuses.

Public Health and Safety Constraints

The system must have stronger encryption to protect against hackers or unauthorized persons trying to access sensitive information from the system. This will ensure the security and safety of users' information.

User Constraints

The system must have a user-friendly interface for non-technical users since it may be a hindrance for non technical users or not familiar with a web based design like faculty members. All users in the main as well as the 13 campuses are required to join a seminar or meeting on the use of the system to ensure the effectiveness and consistent use of the system.

Budget and Resource Constraints

The project system operates within a limited budget, since this is a university-based system. Development obeys a constrained timeline, which has efficient planning and resource allocation. After Deployment, ongoing maintenance and its cost need to be considered to have long term functionality of the system. The system's sustainability depends on the availability of existing technical support staff.

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